

Homework — 1.3.2 Databases

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.3.2 lesson. Show your working.

Part A — This week's topic (~14 marks)

A sports club uses the following two tables. Underlined attributes are primary keys.

Member(MemberID, FirstName, Surname, MembershipType) Booking(BookingID, MemberID, CourtNumber, BookingDate)

1. Identify the **foreign key** in this design and state which table and attribute it links to. **[2]** (AO2)

2. Write an SQL query that returns the FirstName and Surname of every member whose MembershipType is 'Junior', sorted by Surname in **descending** order. **[3]** (AO2)

3. Write an SQL statement that uses a **JOIN** to list each member's Surname together with the CourtNumber and BookingDate of every booking they have made. **[3]** (AO2)

4. Consider the unnormalised data below, where one order can contain several products:

Order(OrderID, CustomerName, CustomerAddress, ProductID, ProductName, Quantity)

(a) Explain why this table is **not** in First Normal Form (1NF) as a single record per order, and state what 1NF requires. **[2]** (AO1/AO2)

(b) `ProductName` depends on `ProductID`, which is not the whole key of the order-line. Name the type of dependency this is and state which normal form removes it. **[2]** (AO1)

5. Explain the **Durability** property of a database transaction. **[2]** (AO1)

Part B — Retrieval: keep it fresh (~6 marks)

6. (1.3.1) A digital photograph is to be uploaded to a website. State whether **lossy** or **lossless** compression is the more suitable choice and give **one** reason. **[2]** (AO1/AO2)

7. (1.2.3) State what is meant by the **Waterfall** model of software development and give **one** drawback of it. **[2]** (AO1)

8. (1.1.3) Explain **one** reason why an SSD might be chosen over a magnetic hard disk drive in a laptop.
[2] (AO1)
